

Python Dictionary Operations, Frequency Analysis & Comprehensions

1. Count Characters Frequency in a String

Aim: To count the frequency of each character in a string.

```
text = "dictionary"
char_count = {}

for char in text:
    if char in char_count:
        char_count[char] += 1
    else:
        char_count[char] = 1

print(char_count)
```

```
{'d': 1, 'i': 2, 'c': 1, 't': 1, 'o': 1, 'n': 1, 'a': 1, 'r': 1, 'y': 1}
```

Result: Thus the given program Count Characters Frequency in a String was successfully verified.

2. Filter Dictionary Items by Value

Aim: To filter dictionary items based on their values.

```
scores = {"A": 4, "B": 68, "C": 98, "D": 40}
filtered_scores = {}

for key, value in scores.items():
    if value > 50:
        filtered_scores[key] = value

print(filtered_scores)
```

```
{'B': 68, 'C': 98}
```

Result: Thus the given program Filter Dictionary Items by Value was successfully verified.

3. Find the Key with Maximum Value

Aim: To find the key having the maximum value in a dictionary.

```
marks = {"Siri": 98, "Vendhan": 100, "Vikki": 88}

max_key = None
max_value = 0

for key, value in marks.items():
    if value > max_value:
        max_value = value
        max_key = key

print(max_key)

Vendhan
```

Result: Thus the given program Find the Key with Maximum Value was successfully verified.

4. Square Each Value in a Dictionary

Aim: To write python code to square each value in a dictionary.

```
nums = {"a": 2, "b": 4, "c": 6}
squared_nums = {}

for key, value in nums.items():
    squared_nums[key] = value ** 2

print(squared_nums)

{'a': 4, 'b': 16, 'c': 36}
```

Result: Thus the given program Square Each Value in a Dictionary was successfully verified.

5. Combine Two Lists into a Dictionary

Aim: To combine two lists into a dictionary.

```
keys = ["id", "name", "age"]
values = [10, "Leo", 20]

combined_dict = {}

for i in range(len(keys)):
```

```
combined_dict[keys[i]] = values[i]
print(combined_dict)
{'id': 10, 'name': 'Leo', 'age': 20}
```

Result: Thus the given program Combine Two Lists into a Dictionary was successfully verified.

6. Count Word Frequency in a Sentence

Aim: To count the frequency of each word in a sentence.

```
sentence = "Playing football is simple, but playing simple football is the hardest"
words = sentence.split()
word_count = {}
for word in words:
    if word in word_count:
        word_count[word] += 1
    else:
        word_count[word] = 1
print(word_count)
{'Playing': 1, 'football': 2, 'is': 2, 'simple': 1, 'but': 1, 'playing': 1, 'the': 1, 'hardest': 1}
```

Result: Thus the given program Count Word Frequency in a Sentence was successfully verified.

7. Swap Keys and Values in a Dictionary

Aim: To swap the keys and values of a dictionary.

```
data = {"a": 1, "b": 2, "c": 3}
swapped_data = {}
for key, value in data.items():
    swapped_data[value] = key
print(swapped_data)
{1: 'a', 2: 'b', 3: 'c'}
```

Result: Thus the given program Swap Keys and Values in a Dictionary was successfully verified.

8. Sum All Values in a Dictionary

Aim: To find the sum of all values in a dictionary.

```
expenses = {"Food": 1000, "Rent": 6000, "Travel": 2400}
total = sum(expenses.values())
print(total)
9400
```

Result: Thus the given program Sum All Values in a Dictionary was successfully verified.

9. Filter Dictionary by Value Using Comprehension

Aim: To filter dictionary items based on their values using dictionary comprehension.

```
scores = {"A": 46, "B": 78, "C": 88, "D": 40}
filtered_scores = {key: value for key, value in scores.items() if
value > 50}
print(filtered_scores)
{'B': 78, 'C': 88}
```

Result: Thus the given program Filter Dictionary by Value Using Comprehension was successfully verified.

10. Square All Values Using Dictionary Comprehension

Aim: To square all values in a dictionary using dictionary comprehension.

```
numbers = {"a": 2, "b": 3, "c": 4}
squared_numbers = {key: value ** 2 for key, value in numbers.items()}
print(squared_numbers)
{'a': 4, 'b': 9, 'c': 16}
```

Result: Thus the given program Dictionary Comprehension was successfully verified.

11. Invert Keys and Values using Dictionary Comprehension

Aim: To invert the keys and values of a dictionary.

```
data = {"a": 1, "b": 2, "c": 3}
inv_data = {value: key for key, value in data.items()}
print(inv_data)
{1: 'a', 2: 'b', 3: 'c'}
```

Result: Thus the given program Invert a Dictionary was successfully verified.

12. Merge Two Dictionaries Keeping Larger Values

Aim: To merge two dictionaries by keeping the larger value for common keys.

```
d1 = {"a": 10, "b": 20, "c": 30}
d2 = {"b": 25, "c": 15, "d": 40}
all_keys = set(d1) | set(d2)
merged = { key: max(d1.get(key, 0), d2.get(key, 0)) for key in all_keys}
print(merged)
{'b': 25, 'd': 40, 'a': 10, 'c': 30}
```

Result: Thus the given program Merge Two Dictionaries Keeping Larger Values was successfully verified.